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Oxygen-limited aeration for relieving the impact of phenolic compounds in anaerobic treatment of coal gasification wastewater

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ABSTRACT

Anaerobic digestion is an efficient tool for improving aerobic biodegradability of coal gasification wastewater, but the shock loading usually results in a failed performance of the anaerobic process. The oxygen-limited aeration was conducted to handle the impact of coal gasification wastewater in an anaerobic expanded-bed GAC reactor. Results indicated that the oxygen-limited aeration of 0–0.5 mgO_2/L could shorten the recovery time from 23 to 11 days under natural condition, and the addition of 1 g/L PAC could reduce the recovery time to 14 days. The corresponding removal efficiencies of COD and phenolic compounds could increase to above 60 and 50%, respectively. The phenol utilization rates of sludge reached 83.9 ± 5.7 mg phenol/(g VSS d) after recovery of reactor performance by using oxygen-limited aeration, and only 70.4 ± 3.8 mg phenol/(g VSS d) by using natural recovery and 65.2 ± 5.9 mg phenol/(g VSS d) by using PAC addition. The recovery strategy of oxygen-limited aeration might be helpful to relieve the inhibitory effect of phenolic compounds and restore metabolic activity of anaerobic bacteria.

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1. Introduction

At present, in China, coal syngas is produced in large quantities to meet the requirement of natural gas and to provide the raw materials in the manufacture of chemical products, such as methanol, synthetic ammonia and olefin (Chen and Xu, 2010; Yoon and Lee, 2012). However, the difficulties in treatment of coal gasification wastewater (CGW) are far beyond the imagination. The CGW contained phenolic compounds, heterocyclic and polycyclic aromatic hydrocarbons, long chain alkanes, ammonia, thiocyanate, and cyanide (Sharma et al., 2012). Among these compounds, phenolic compounds were the main organic pollutants and contributed about 30–60% of the COD content (Wang et al., 2010). Especially, the concentration of total phenols was still around 400 mg/L even though the pretreatment of solvent extraction (Gai et al., 2008; Yu et al., 2010). The toxicity and recalcitrance of phenolic compounds inhibited its biological degradation. Generally, the CGW was treated using anoxic–oxic process or conventional activated sludge process, because the most of phenolic compounds could be degraded readily under aerobic conditions (Li et al., 2011; Kim et al., 2011a,b). However, the aerobic biological treatment could not reduce the concentrations of COD and phenolic compounds in the CGW to meet the national emission standards.

Many attentions had paid to anaerobic biological technologies in the treatment of CGW because of their advantages, which included low energy consumption, detoxification and improvement of biodegradability (Fox et al., 1990; Nakhla et al., 1990; Wang et al., 2011). In the 1980s of last century, anaerobic activated carbon filter was used to treat the CGW, while excellent removal efficiency of organic matter was achieved by a combination of anaerobic fermentation and adsorption (Suidan et al., 1983). Operational techniques using expanded-bed or fluidized-bed anaerobic granular activated carbon (GAC) bioreactors were developed subsequently for reducing the cost in treating the CGW (Pfeffer and Suidan, 1989; Fox et al., 1990; Nakhla et al., 1990). Because of the toxicity of CGW, the full strength wastewater was not treatable at organic loading rates (OLR) above 10 kg COD/(m^3 d) while the GAC reactors treating the diluted wastewater could achieve around 90% removal of COD and over 99.5% reduction of phenolic compounds at OLR range of 32–34 kg COD/(m^3 d) in the synthetic CGW (Nakhla and Suidan, 1995). Furthermore, the periodic partial replacement of GAC was also an effective way to overcome the toxicity of the wastewater in the anaerobic GAC reactors (Pfeffer and Suidan, 1989; Nakhla et al., 1990). Except the anaerobic GAC technologies, anaerobic treatment of actual CGW was investigated in the conventional anaerobic processes, such as two-continuous UASB

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reactors and addition of PAC in the UASB reactor (Wang et al., 2011; Wang and Han, 2012). These results demonstrated that anaerobic digestion had a significant improvement effect on the biodegradability of CGW, but the COD removal exceeding 70% was difficult at OLR range of 2.5–3.0 kg COD/(m^3 d) in the absence of activated carbon. Obviously, anaerobic expended-bed GAC (EB-GAC) reactor was more efficient than the conventional anaerobic processes for treating the actual CGW.

In fact, the composition of actual CGW was complicated and varied from one gasification plant to another, depending on the quality of raw coal, gasifier types, gasification temperature, and recovery methods of byproducts. So, the application of anaerobic processes in the treatment of CGW had been hampered by a lack of understanding of factors associated with process stability, such as shock loading and recovery methods. In the anaerobic GAC reactors, one method using the periodic partial replacement of GAC had been successfully applied for reducing the toxicity and enhancing the stability in the laboratory studies (Nakhla et al., 1990). However, on an industrial scale, the shock loading appeared frequently in the influent, and the periodic partial replacement of large volumes of GAC was unreasonable and very costly.

The aim of this study was to investigate the effect of oxygen-limited aeration and addition of powdered activated carbon (PAC) on the recovery process of anaerobic EB-GAC reactor treating CGW after shock loading. The start-up of the anaerobic EB-GAC reactor was recorded and the impact of shock loading was evaluated on the operation of the EB-GAC reactor. Comparing with the natural recovery, the recovery process and the mechanism of oxygen-limited aeration and addition of PAC were analyzed.

2. Material and methods

2.1. Experimental setup and inoculum

The EB-GAC reactor was constructed of cylindrical plexiglass column with an internal diameter of 5 cm and a working volume of 1 L. A supporting layer was placed at the bottom of the reactor to support the GAC and distribute influent wastewater and air. The temperature was maintained at 35 ± 2 °C by a constant temperature controller through electric heating. The reactor was initially charged with 100 g of 16 × 20 mesh GAC. An internal recycle wastewater of 0.2 L/min provided an initial expansion of GAC bed in the reactor. The seed sludge was obtained from a full-scale anaerobic reactor at a CGW treatment plant in the northeast region of China. The initial suspended solids (SS) and volatile suspended solids (VSS) added to the reactor were around 10.5 and 7.1 g/L, respectively.

2.2. Wastewater characteristics and GC-MS analysis

The organic composition of CGW after pretreatment by ammonia stripping and phenol extraction with methyl isobutyl ketone (MIBK) was presented in the previous study (Wang et al., 2011). The organic composition of GC-MS analysis is shown in Fig. 1. It was not difficult to find that phenolic compounds, such as phenol, *p*-cresol and 2-[(trimethylsily)oxy]-phenol, were the major organic constituents of the CGW. The actual CGW had a complicated composition and the wastewater characteristics with their constituent concentrations expressed in (mg/L) were as follows: COD (2023 ± 146), BOD_5 (685 ± 47), phenolic compounds (392 ± 35), volatile fatty acids (759 ± 84), ammonia nitrogen (129 ± 10), oil (45 ± 6), thiocyanate (65 ± 7) and pH (7.5 ± 0.3). The macro- and micro-nutrients were added to the anaerobic EB-GAC reactor as described in the early work (Wang et al., 2010).

2.3. Start-up and operation

The operational schedule of the EB-GAC reactor was divided into four phases: start-up, natural recovery, addition of PAC, and oxygen-limited aeration. Normal operation of the EB-GAC reactor was maintained at COD and phenolic concentrations of around 2000 and 400 mg/L at a HRT of 24 h. The first phase lasted for 156 days during the start-up period. For the second period extending to day 194, the organic loading on the EB-GAC reactor was maintained at 5.16 kg COD/(m^3 d) with 24 h between days 157 and 158, while the concentrations of COD and phenolic compounds increased from 1859 and 354.5 mg/L to 5160 and 815 mg/L respectively in the influent. The anaerobic EB-GAC reactor was allowed to recover fully from the shock loading and the recovery time was defined as the time between the initiation of recovery action and the time when the removal efficiency of phenolic compounds achieved above 50%. After the natural recovery of reactor performance, the EB-GAC reactor was continuously operated at a pseudo-steady state for two weeks. Between days 195 and 196, the shock loading of 5.32 kg COD/(m^3 d) was exerted on the EB-GAC reactor for one day time. 1 g

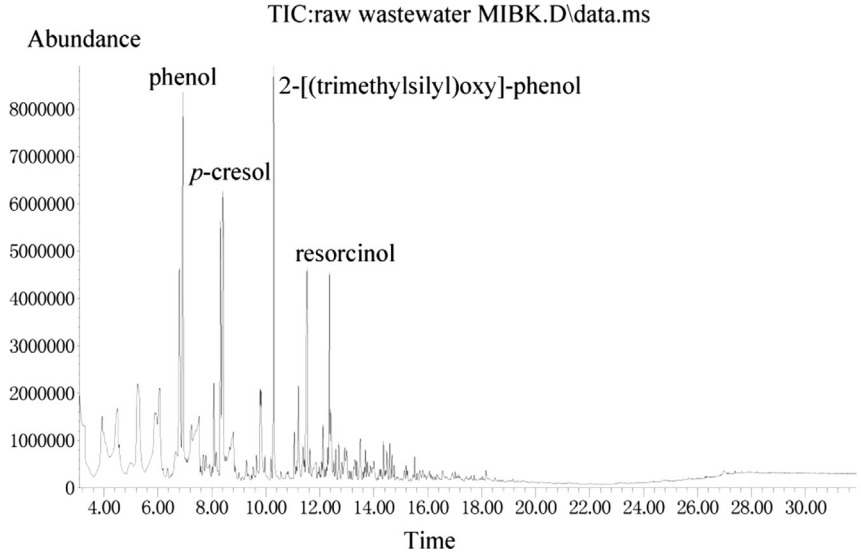


Fig. 1. Main organic composition in the influent by GC-MS analysis.

PAC per liter wastewater was added in the EB-GAC reactor from days 196 to 199 and then the reactor performance was gradually recovered without the addition of PAC. For the fourth period, the organic loading rate of 5.46 kg COD/(m^3 d) and phenolic loading rate of 0.85 kg/(m^3 d) in the EB-GAC reactor were maintained for 24 h between days 224 and 225. The oxygen-limited aeration of 0–0.5 mg/L was used to the recovery of the reactor performance for 3 days, and then the EB-GAC reactor performance was gradually recovered without aeration on day 227.

2.4. Analytical methods

COD, BOD_5, SS, VSS, phenols, oil, thiocyanate, and ammonia nitrogen were determined according to Standard Methods (Wei et al., 2002). pH values were analyzed by a pH meter (pHS-3C, Leici, China). Volatile fatty acids (VFAs) were analyzed by the distillation method (Anderson and Yang, 1992). The biogas production of the reactor was detected by the gas flowmeter. Methane content of the biogas was measured using a liquid displacement system with a 3 mol/L NaOH solution. Phenol utilization rate (PUR) tests were used to evaluate the metabolic activity of the biomass for degrading phenol before and after the shock loading in the EB-GAC reactor (Wang et al., 2010). For organic composition, the influent was analyzed via GC-MS and the analytical conditions were described in the previous paper (Wang et al., 2002). The morphology of the sludge samples was analyzed using a scanning electron microscope (SEM, S-4700, HITACHI).

3. Results

3.1. Start-up of the EB-GAC reactor

The removal efficiencies of COD and phenolic compounds during the start-up period in the EB-GAC reactor are shown in Fig. 2. For the first four days, both the removal efficiencies of COD and phenolic compounds were above 75%, but subsequently the removal efficiencies decreased quickly to below 15% after day 7. Obviously, the adsorption capacity of GAC was exhausted and the phenolic compounds almost penetrated the EB-GAC reactor. With the influent COD and phenolic compounds of around 2006 and 401 mg/L, the concentrations of COD and phenolic compounds were still above 1000 and 300 mg/L in the effluent even after another 70 days' operation, and the corresponding removal efficiencies of COD and phenolic compounds were mostly below 45 and 20%, respectively. Although the seed sludge had been acclimated to the CGW for a long time under methanogenic conditions, the sludge still needed sufficient time to adapt the environment in the EB-GAC reactor. At this stage, the maximum production of methane was only about 10 mL and no methane production was detected at the most of time. This indicated that the methanogenic organisms were being affected by the toxic substances in the influent. At the end of the start-up period (from days 143 to 156), the removal efficiencies of COD and phenolic compounds remained at around 69 and 52%, and the concentrations of COD and phenolic compounds decreased to around 613 and 195 mg/L, respectively.

3.2. Natural recovery of the EB-GAC reactor after the shock loading

On day 157, the influent COD concentration was increased from 1859 to 5160 mg/L with an increase in the concentration of phenolic compounds from 354.5 to 815 mg/L, while keeping this state for 24 h. Results of the shock loading and natural recovery of the EB-GAC reactor are illustrated in Fig. 3. In the initial days (days 158–159), the concentrations of effluent COD and phenolic compounds suddenly rose to exceed the influent concentrations due to the residual organic contaminants and the increase in effluent suspended solids. The recovery process was slow from days 163 to 180, and the removal efficiencies of COD and phenolic compounds were around 63 and 53% respectively on day 180. The whole time lasted 23 days for the recovery process of the reactor performance. Subsequently,

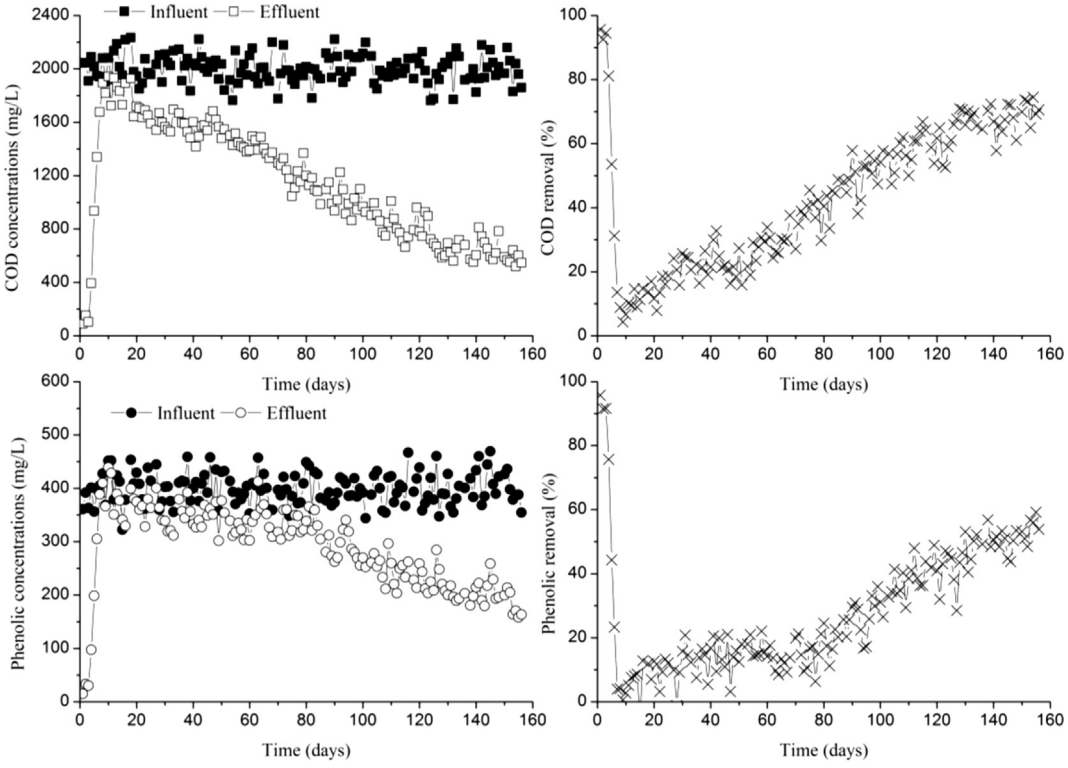


Fig. 2. Removal of COD and phenolic compounds during the start-up period in the EB-GAC reactor.

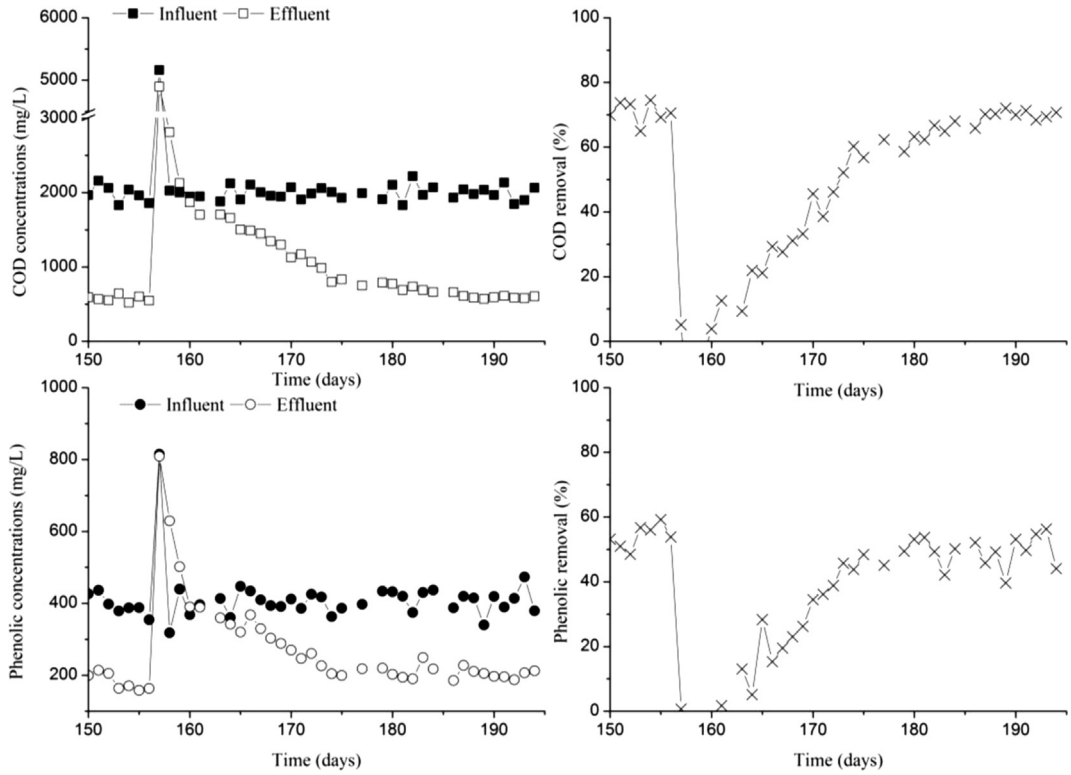


Fig. 3. Results of the shock loading and natural recovery of the EB-GAC reactor.

the EB-GAC reactor was continued with another 14 days' operation. Between days 181 and 194, the removal efficiencies of COD and phenolic compounds reached about 69 and 49%, and the corresponding concentrations were around 627 and 205 mg/L, respectively.

3.3. Recovery after the shock loading by addition of PAC in the EB-GAC reactor

On day 195 of reactor operation, the COD concentrations were increased from 2065 to 5320 mg/L, and the concentration of phenolic compounds increased from 379.6 to 795.7 mg/L. Results of addition of PAC for tackling the shock loading in the EB-GAC reactor are shown in Fig. 4. These changes caused a decrease in COD removal from 70.7 to 6.9% in one day time. In order to shorten the recovery time of reactor performance, 1 g/L of PAC was added into the reactor for adsorbing the pollutants in the influent from days 196 to 198, which reduced the concentrations of COD and phenolic compounds to 1427.5 and 248.2 mg/L respectively in the effluent. This resulted in a significant improvement in treatment efficiency of the anaerobic EB-GAC reactor. Within the next 11 days, the removal efficiencies of COD and phenolic compounds increased to 62.7 and 50.3%, respectively. The recovery time was 14 days after the shock loading in the EB-GAC reactor. To improve the reactor performance, the EB-GAC reactor was continued with two weeks' operation. On day 223, the removal efficiencies of COD and phenolic compounds were 70 and 50.6% and the corresponding concentrations were 574.8 and 219.8 mg/L respectively in the effluent.

3.4. Recovery after the shock loading by oxygen-limited aeration in the EB-GAC reactor

Results of oxygen-limited aeration for tackling the shock loading in the EB-GAC reactor are shown in Fig. 5. On day 224, the shock loading was provided with COD of 5460 mg/L and phenolic compounds of 850 mg/L, and the removal efficiencies of COD and phenolic compounds decreased from 70 and 50.6% to 4.2 and 3.9%, respectively. The oxygen-limited aeration of 0–0.5 mgO_2/L as a recovery strategy was used to accelerate the recovery of reactor performance for 3 days. The effluent concentrations of COD and phenolic compounds were 757 and 129.4 mg/L, respectively on day 227. On day 228, the removal efficiencies of COD and phenolic compounds quickly decreased to 35.2 and 25.8% due to the stop of oxygen-limited aeration. Subsequently, the gradual increase of removal efficiencies was observed in the next 7 days. The removal efficiencies of COD and phenolic compounds were 62.5 and 51.6%, respectively on day 235. As calculated, the recovery time was 11 days after the third shock loading in the EB-GAC reactor. For the next 16 days the influent concentrations of COD and phenolic compounds varied in the range of 1794.5–2224.5 and 350.3–465 mg/L respectively in the EB-GAC reactor. The removal efficiencies of COD and phenolic compounds were about 67.8 and 50.3% respectively during this period.

3.5. Phenol utilization rate tests

PUR tests evaluated the metabolic activities of suspended sludge for degrading phenol in the EB-GAC reactor during the different periods, as shown in Table 1. At the end of the start-up period, the PUR values increased from the initial values of 35.6 ± 5.2 mg phenol/(g VSS d) in the seed sludge to 63.1 ± 7.4 mg phenol/(g VSS d) in the suspended sludge on day 155. When the shock loading happened, the SUR decreased to 43.5 ± 6.0 mg phenol/(g VSS d) on day 160 and recovered to 70.4 ± 3.8 mg phenol/(g VSS d) after the natural recovery on day 193. On day 198, the SUR values reduced to 41.4 ± 3.5 mg phenol/(g VSS d) due to the second shock loading. After the recovery with addition of PAC, the SUR values recovered to 65.2 ± 5.9 mg phenol/(g VSS d). When the third

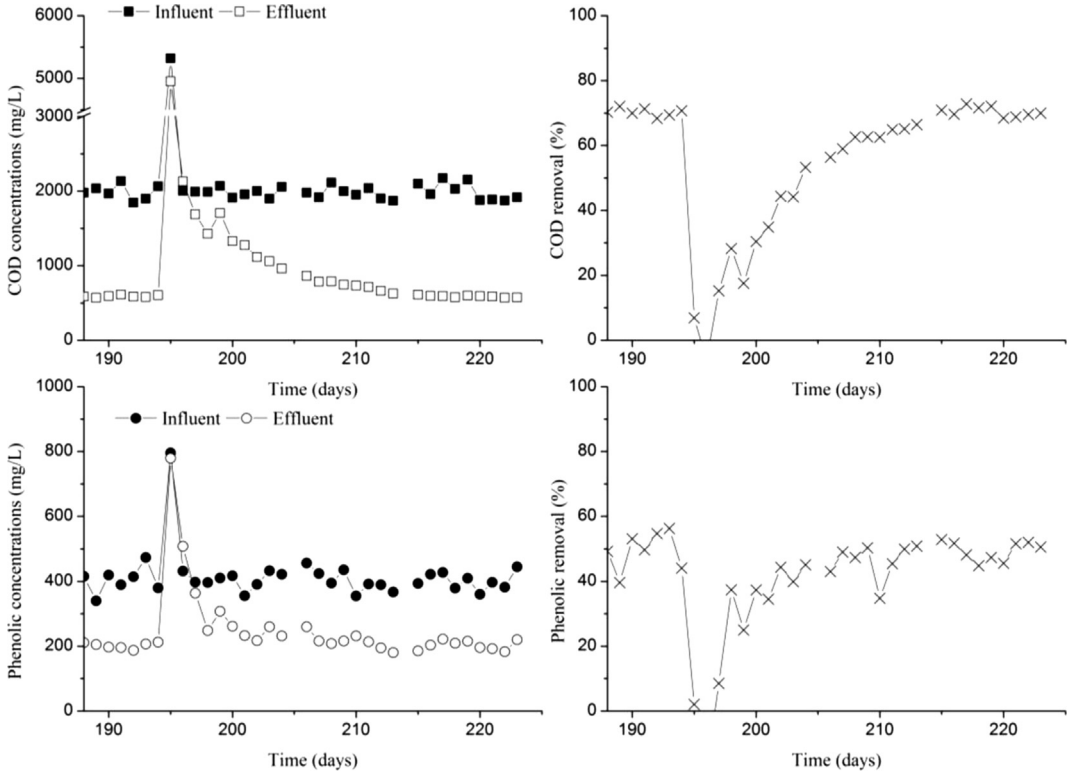


Fig. 4. Results of addition of PAC for tackling the shock loading in the EB-GAC reactor.

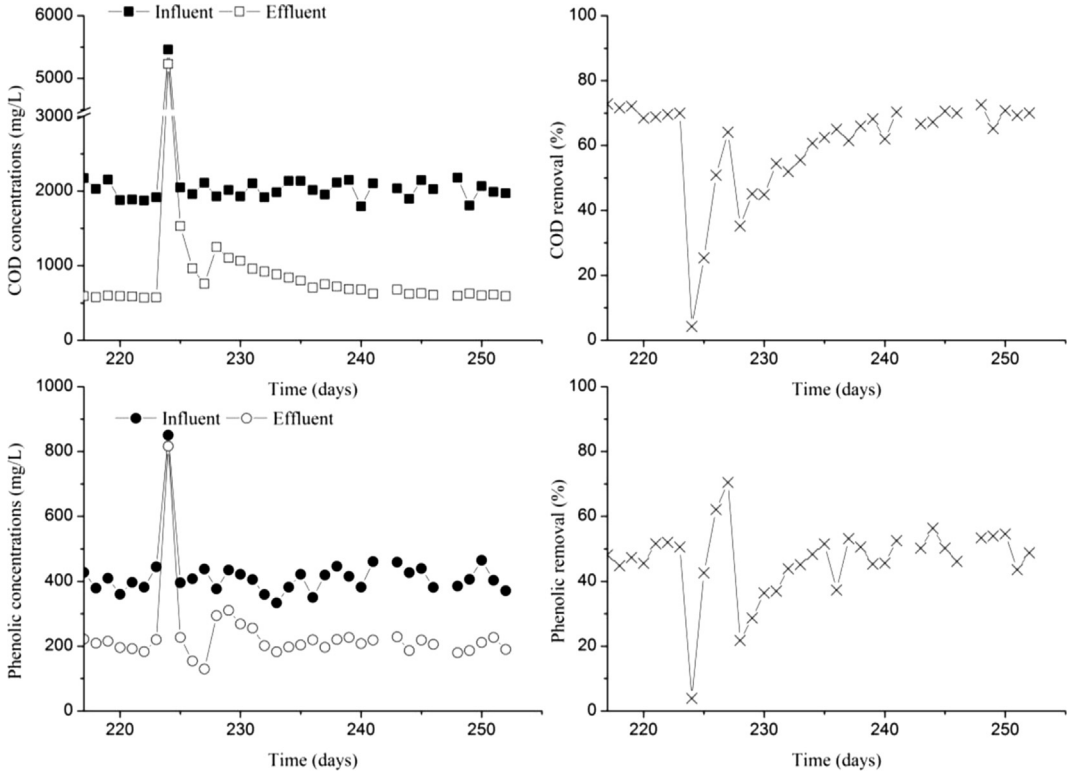


Fig. 5. Results of oxygen-limited aeration for tackling the shock loading in the EB-GAC reactor.

Table 1
PURs of the suspended sludge in the EB-GAC reactor

Different periods	Days	PUR values mg phenol/(g VSS d)
Control	1	35.6 ± 5.2
Start-up	155	63.1 ± 7.4
Natural recovery	160	43.5 ± 6.0
	193	70.4 ± 3.8
PAC addition	198	41.4 ± 3.5
	222	65.2 ± 5.9
Oxygen-limited aeration	227	61.8 ± 4.3
	251	83.9 ± 5.7

shock loading happened, the SUR values were 61.8 ± 4.3 mg phenol/(g VSS d) on day 227 and increased to 83.9 ± 5.7 mg phenol/(g VSS d) on day 251.

3.6. Morphology of the sludge

Fig. 6 shows the photomicrographs from a SEM analysis of the suspended sludge in the reactor and the biofilm on the GAC particles thus indicating that the PAC acted as a support for biofilm growth. The sludge was mainly comprised of short-rod type bacteria (Fig. 6a and b). The photomicrographs showed the dominance of short-rod type bacteria embedded by a thick layer of extracellular polymeric substances (EPS) (Fig. 6a). It was reported that a mass of EPS would be produced when the anaerobic sludge suffered from toxic environment (Aquino and Stuckey, 2004). On the basis of appearance, the biomass in the reactor was mainly composed of acidogenic-like bacteria and the predominant bacteria which were similar from that of biofilm in GAC surface.

4. Discussion

According to the previous studies, anaerobic digestion plays a positive role in the biological treatment of coal gasification wastewater (Wang et al., 2011). With the influent COD and phenolic compounds of around 2000 and 400 mg/L, the maximum removal efficiencies can reach above 70 and 50% respectively at the HRT of 24 h in the EB-GAC reactor. Researchers have reported that successful operation of the anaerobic GAC system requires a carbon replacement schedule to be maintained, which can keep the bulk concentrations of toxic compounds below their toxic threshold (Nakhla et al., 1990). Before the shock loading, efficient COD removal and phenolic degradation were achieved in the EB-GAC reactor, which confirmed the feasibility of this wastewater treatment approach. However, due to the inability to timely replacement of GAC, the EB-GAC reactor would also run to failure once the shock loading impacted and the recovery time was long without any measures. So, the instability of the EB-GAC reactor was still the most disadvantages for its application in the treatment of CGW.

As known, as long as the toxic substances can be absorbed by the activated carbon, they would not inhibit the anaerobic bacteria. But, the adsorption capacity of GAC or PAC was very low for the actual CGW due to its high concentration and complex composition. If there were high concentrations of toxic compounds in the influent, the recovery strategies should reduce these toxic compounds to the concentrations below their threshold levels. Previous works had indicated that addition of PAC and oxygen-limited aeration was the best recovery strategies in the UASB reactor (Wang and Han, 2012). Also, in the EB-GAC reactor, addition of PAC of 1 g/L shortened the recovery time from 23 to 14 days and oxygen-limited aeration of 0–0.5 mgO_2/L reduced the recovery time to 11 days. Obviously, these two recovery strategies could shorten the recovery time after shock loading, especially oxygen-limited aeration. In fact, the limited aeration had been employed as an efficient way to achieve high treatment efficiency under anaerobic conditions (Zitomer and Shrout, 1998; Zhou et al., 2007; Xu et al., 2014). The recovery strategy of oxygen-limited aeration was more suitable for the EB-GAC reactor than the UASB reactor, which the aeration didn't result a huge loss of sludge in the EB-GAC reactor.

Phenolic compounds didn't accumulate in the EB-GAC reactor when the GAC attained adsorption saturation. When the bacteria utilized the phenolic compounds that adsorbed by the GAC, the biological GAC reactor might played an important role in degradation of phenolic compounds. After the shock loading, the high concentration of phenolic compounds not only caused rapid depletion of adsorption capacity of GAC, but also inhibited seriously the activity of anaerobic sludge. Addition of PAC could reduce the bioavailable concentration of phenolic compounds to tackle the shock loading, but the recovery rate of the activity of anaerobic sludge was relatively slow. It was noteworthy that oxygen-limited aeration could effectively reduce the residual concentration of phenolic compounds and also could quickly recover the activity of anaerobic sludge. The degradation rates of phenolic compounds under oxygen-limited condition were higher than the values under anaerobic condition. In the traditional sense, the methanogenic microbes were sensitive to toxic pollutants. However, the metabolic activity of anaerobic sludge wasn't inhibited completely by the emergence of molecular oxygen (Zitomer and Shrout, 1998). After

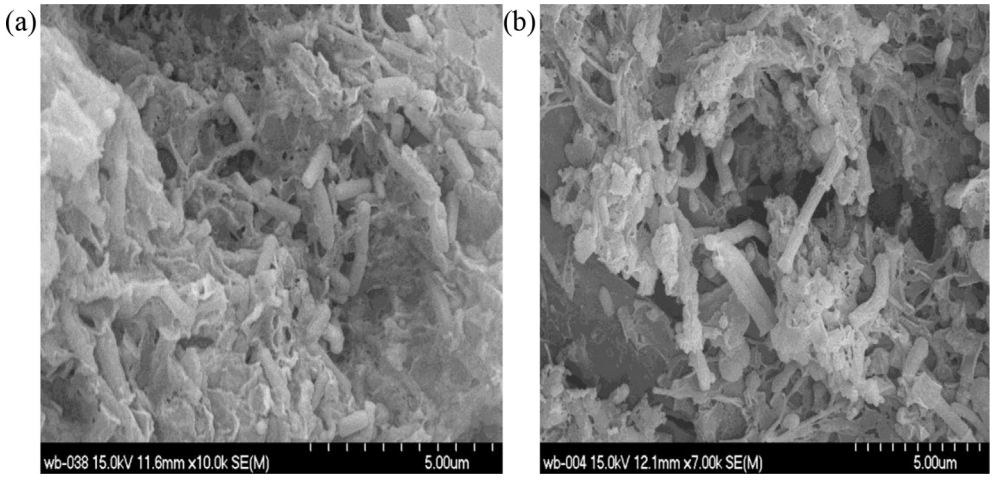


Fig. 6. Photomicrographs from a SEM analysis of biomass: (a) suspended sludge from the reactor; (b) biofilm of the GAC particles.

the recovery strategy of oxygen-limited aeration, the methane production was still observed with a short-term adaption, the total gas production was lower than expected values from the theoretical calculation. So, it was speculated that the anaerobic microorganisms were mainly composed of facultative bacteria in the EB-GAC reactor. In addition, the EPS was considered to be a defensive strategy to protect microbes against shock loadings caused by toxic compounds, because the concentration of toxic compounds in the extracellular was generally lower than in the intracellular due to the diffusion effects. Comparing with addition of PAC, the recovery strategy of oxygen-limited aeration might be helpful to relieve the inhibition caused by toxic pollutants and restore the metabolic activity of anaerobic microbes. Thus, oxygen-limited aeration was a promising strategy to handle the shock loading in anaerobic treatment of CGW. It was suggested that the potential application of this strategy was suitable for anaerobic treatment of CGW. Further research is required to establish the optimum conditions of recovery strategies.

5. Conclusion

The EB-GAC process provided an efficient technique for anaerobic treatment of coal gasification wastewater, but the shock loading of around 5.2 kg COD/(m^3 d) would result in a failed performance. The PAC addition and oxygen-limited aeration could shorten the recovery time from 23 days under natural condition to 14 and 11 days, respectively. As compared to addition of PAC, the oxygen-limited aeration might be helpful to reduce microbial inhibition caused by toxic pollutants and restore metabolic activity of anaerobic bacteria. The oxygen-limited aeration was a promising strategy to handle the shock loading in anaerobic treatment of CGW.

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